Q.1

$$A = \begin{bmatrix} x & \frac{x}{y} & 0 \\ e^{-x} & -\frac{1}{y} & \sqrt[3]{x} \\ \sin\theta & -3 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

Q.2

Sales Data				
Seasons	Salesman			
	A	B	C	D
Summer	36	21	35	35
Winter	28	29	31	32
Monsoon	26	28	29	29

Q.3

The arc length of the curve $y = f(x)$ over the interval $a \leq x \leq b$ is $\int_a^b \sqrt{1 + [f'(x)]^2}\, dx$. For the function $f(x) = x^3$, $0 \leq x \leq 1$, find the approximate arc length correct to four decimal places using Trapezoidal rule with $h = 0.1$.

Q.4

(a) Descriptive Research

(b) Descriptive Research

(c) Descriptive Research

(d) Descriptive Research

(e) *Descriptive Research*

Q.5

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right).$$

Q.6

$$f(t) = \begin{cases} 0, & t < 0, \\ e^{-t}, & t > 0. \end{cases}$$

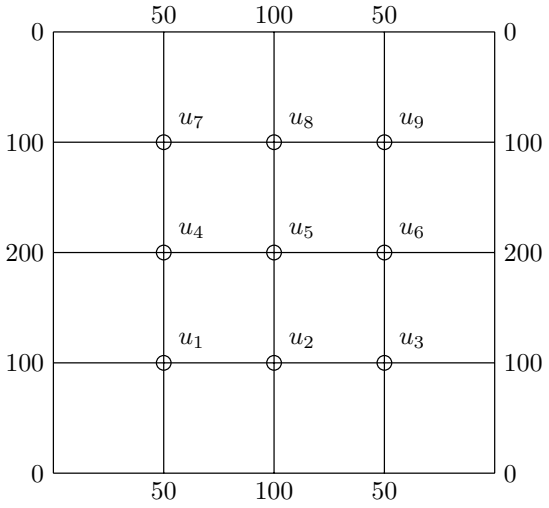
Demo Question Paper
Differential Equations

Q.2 (a) Use suitable Fourier transform to solve the following partial differential equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{4}\frac{\partial^2 u}{\partial t^2}, \quad -\infty < x < \infty,\ t > 0,$$
$$u(x,t) \to 0 \text{ as } |x| \to \infty,$$
$$u(x,0) = 0, \quad u_t(x,0) = e^{-|x|}, \quad -\infty < x < \infty.$$

Q.8

Q.8. Use Liebmann's iteration formula to solve Laplace equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ at the interior mesh points of the square region with boundary values given in the following figure.



[Take 2 iterations and obtain the result correct up to three decimal places.]

Q.9

Quantum Mechanics

Motivation in Research

Nikhil S Sulghur

Mumbai University

August 15, 2026

Q.9 Axioms of Quantum Mechanics

1. **State:** A system is described by a normalized state vector $|\psi\rangle$ in a Hilbert space.
2. **Observables:** Every measurable quantity corresponds to a Hermitian operator $\hat{A}$.
3. **Measurement:** The only possible results are the eigenvalues a_n of $\hat{A}$, with probability $|\langle a_n|\psi\rangle|^2$.
4. **Collapse:** After measuring a_n, the state becomes the eigenstate $|a_n\rangle$.
5. **Evolution:** $i\hbar \dfrac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle$.

References


D. J. Griffiths, *Introduction to Quantum Mechanics*, Pearson.

Thank You

I thank the Department of Mathematics for this opportunity.

(a) Text style and emphasis

Bold, *italic*, SMALL CAPS, `typewriter`, underlined, *emphasised*, red.

(b) Text sizing

tiny scriptsize footnotesize small normalsize large Large LARGE huge Huge

(c) Paragraph alignment and environments

This line is centred.

This line is left aligned.

This line is right aligned.

(d) Document structure

A subsection

A subsubsection

This page uses `\documentclass`, `\usepackage`, `\section*` and the `document` environment.

(e) Lists and layout

- Apple
- Mango

1. First
2. Second

---

1 Introduction

This is the first page. The header and footer set above appear on every page.

2 Next Section

This is the second page, showing the page number change in the footer.

Q.12

Nikhil S Sulghur
M.Sc.-I, Department of Mathematics
Mumbai University
October 18, 2026

The Head of the Department
Department of Mathematics
Mumbai University

Respected Sir/Madam,

Subject: Request for permission to use MATLAB

I am a student of M.Sc.-I, Department of Mathematics. I kindly request your permission to use MATLAB in the departmental computer laboratory for my research project work.

I shall use the software only for academic purposes and will follow all laboratory rules. I would be grateful for your kind approval.

Yours faithfully,

Nikhil S Sulghur

Research in Mathematics: Real Analysis

Nikhil S Sulghur

October 18, 2026

Declaration

I hereby declare that this project report is my own work and has not been submitted to any other university or institution for any degree or diploma. All sources used are acknowledged in the bibliography.

Acknowledgement

I express my sincere gratitude to the Department of Mathematics, Mumbai University, for giving me this opportunity. I also thank my teachers and friends for their constant guidance and support.

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3.3 Examples 6

4 Continuity **7**
4.1 Introduction 7
4.2 Main Content 7
4.3 Examples 7

5 Conclusion **8**

Chapter 1

Number Systems

1.1 Introduction

This chapter introduces the real number system. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

1.2 Main Content

We study ordered sets and the least upper bound property. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

1.3 Examples

The rational numbers do not have this property. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 2

Basic Topology

2.1 Introduction

This chapter introduces metric spaces. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

2.2 Main Content

We study open sets and compact sets. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

2.3 Examples

Closed and bounded intervals are compact subsets of the real line. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 3

Sequences and Series

3.1 Introduction

This chapter introduces numerical sequences. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

3.2 Main Content

We study convergence and Cauchy sequences. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

3.3 Examples

The sequence $1/n$ converges to zero. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 4

Continuity

4.1 Introduction

This chapter introduces limits of functions. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

4.2 Main Content

We study continuous functions on metric spaces. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

4.3 Examples

Polynomial functions are continuous on the whole real line. This topic is discussed in Rudin [1] and forms a basic part of real analysis.

Chapter 5

Conclusion

In this report we studied the basic ideas of real analysis, namely number systems, topology, sequences and continuity, following Rudin [1]. These ideas form the foundation for further study of differentiation and integration.

Bibliography

[1] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976.

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Existence and Uniqueness of Solutions of the Second Order Differential Equations

Nikhil S Sulghur
Mumbai University

October 18, 2026

Abstract

We study the existence and uniqueness of solutions of second order differential equations. We state the Picard–Lindelöf type result for the initial value problem and give a simple example.

1 Introduction

Second order differential equations arise in many problems of physics and engineering. A basic question is whether a solution exists and whether it is unique; see Tsamatos and Ntouyas [1] for boundary value problems of this type.

2 Preliminaries

Consider the initial value problem

$$y'' = f(x, y, y'), \quad y(x_0) = y_0, \quad y'(x_0) = y_1.$$

Setting $z = y'$ turns it into a first order system, so results for systems can be used.

3 Main Result

Theorem. If f is continuous and Lipschitz continuous in y and y' on a rectangle around (x_0, y_0, y_1), then the problem has a unique solution on some interval around x_0.

4 Example

For $y'' + y = 0$ with $y(0) = 0$ and $y'(0) = 1$, the function f is linear, so the theorem applies. The unique solution is $y = \sin x$.

5 Conclusion

Under a Lipschitz condition the initial value problem for a second order equation has exactly one solution. This idea extends to more general boundary value problems.

References

[1] P. Ch. Tsamatos and S. K. Ntouyas, Existence and uniqueness of solutions of a general type boundary-value problem for 2nd-order functional-differential equations, *J. Math. Anal. Appl.*, Elsevier, 1993.

Preface

This page is numbered i.

1 Introduction

This page is numbered 1.

2 Main Content

This page is numbered 10, and the next page will be 11.

Figure 1 shows an inserted picture.

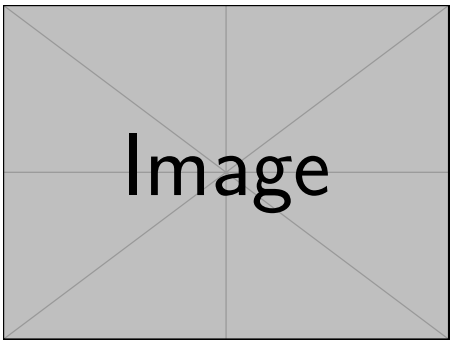


Figure 1: A sample figure

Itemize (bullets):

- Apple
- Mango

Enumerate (numbers):

1. Apple
2. Mango

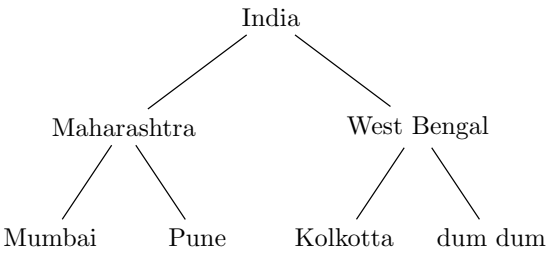
CERTIFICATE

This is to certify that Mr./Ms. _____, Roll No. _____, a student of M.Sc.-I, Department of Mathematics, Mumbai University, has successfully completed the project report titled "_____ during the academic year _____, under my guidance and supervision.

Date: _____ Project Guide: _____

Place: _____ Head of the Department: _____

Q.19


India
Maharashtra
West Bengal
Mumbai
Pune
Kolkotta
dum dum

Q.20

Day	Attendance
Monday	57
Tuesday	11
Wednesday	96
Thursday	122
Friday	210
Saturday	198
Sunday	40
Total	724